

# **RMO– 2014(1)**

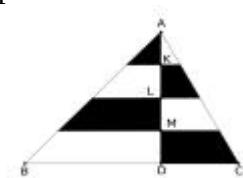
Max Mark – 102

Time – 3 Hrs

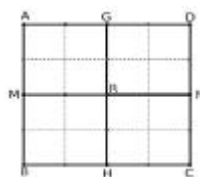
Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions. All Questions carry equal marks.
- Answer to each questions starts on a new page. Clearly indicate the question number.

1. Let ABC be a triangle and let AD be the perpendicular from A on to BC. Let K, L, M be points on AD such that  $AK = AL = LM = MD$ . If the sum of the areas of the shaded regions is equal to the sum of areas of the unshaded regions, prove that  $BD = DC$ .



2. Let  $a_1, a_2, a_3, \dots, a_{2n}$  be arithmetic progression of positive real numbers with common difference  $d$ . Express  $d$  in terms of  $x, y$  and  $z$ , where
  - (i)  $a_1^2 + a_3^2 + \dots + a_{2n-1}^2 = x$
  - (ii)  $a_2^2 + a_4^2 + \dots + a_{2n}^2 = y$
  - (iii)  $a_n + a_{n+1} = z$ .
3. Suppose for some integers  $r$  and  $s$ , the digits of  $2^r$  is obtained by permuting the digits of  $2^s$  in decimal expansion. Prove that  $r = s$ .
4. Is it possible to write the numbers 17, 18, 19, ..., 32 in a  $4 \times 4$  grid of unit squares, with one number in each square, such that product of the numbers in each  $2 \times 2$  sub-grids AMRG, GRND, MBHR and RHCN is divisible by 16?



5. Let ABC be an acute angled triangle and let H be its ortho-centre. For any point P on the circum-circle of triangle ABC, let Q be the point of intersection of the line BH with the line AP. Show that there is a unique point X on the circum-circle of ABC such that for every point  $P \neq A, B$ , the circum-circle of HQP pass through X.
6. Let  $x_1, x_2, x_3, \dots, x_{2014}$  be positive real numbers such that  $\sum_{j=1}^{2014} x_j = 1$ . Determine with proof the

smallest constant K such that  $K \sum_{j=1}^{2014} \frac{x_j^2}{1 - x_j} \geq 1$ .

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